

This section is from the NRA publication "Handloading" book and article "Some Simplified Interior Ballistics for Handloaders" by WM. C. DAVIS, JR. The purpose of this page is not to reprint the article but to give the how for the math nuts out there like me. This book is out of print but might still be found through dealers that specialize in out of print books, such as "Rutgers Book Center (Tel: 732-545-4344)." I would trust that you buy this book if at all possible, for all the great articles. Reprinted in part with the permission of Mr. WM. C. DAVIS, JR.

Special
Notice:

The article states clearly that interior-ballistic calculations alone are not to be trusted to establish safe handloading data. In the interests of safety, the handloader must check all loads against a reliable source of pressure-tested loading data, such as the data published by powder companies and bullet manufacturers, before attempting to assemble and fire such loads.

There are several important characteristics in the performance of a cartridge, these are the charge weight, the capacity of the powder chamber, the diameter and weight of the bullet, and the length of the barrel. When these are combined in various ways you can derive loading density, expansion ratio, sectional density, and the ratio of charge weight to bullet weight. You will find it convenient to record the following measurements by their descriptive name and identify them by the alphabetic variable names listed in the "List of Variables" below. For this section we are going to dispense with the traditional mathematical variables and we will create our own to be more user friendly.

LIST OF VARIABLES:

- A = Mass ratio (charge weight/bullet weight)
- B = Bullet length (in inches)
- C = Case length (in inches)
- D = Bullet diameter (in inches)
- E = Barrel length (in inches)
- F = Full water capacity of case (in grains)
- G = Bullet weight (in grains)
- H = Height (axial length) of boattail (in inches)
- I = Charge weight (in grains)
- J = Tail diameter (small end) of boattail (in inches)
- K = Displacement correction for boattail bullet (in grains)
- L = Cartridge overall length (in inches)
- LD = Density of loading ($I \div W$ ratio)
- M = $1 \div$ the fourth root of R
- N = $(1 - M)$
- P = Water displaced by seating flat-base bullet (in grains)

- Q = Effective bore volume (cubic inches)
- R = Expansion ratio
- S = Bullet seating depth (in inches)
- T = Bullet travel in barrel (in inches)
- U = Volume of cartridge powder chamber (in cubic inches)
- V = Muzzle velocity (ft/sec)
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- ~~X = Powder selection index~~ W = Water capacity of cartridge powder chamber (in grains)
- Y = (G + I ÷ 3)
- Z = Bullet sectional density (1 lb/in²)

Powder space:

The corner stone of interior ballistics is the amount of space available for the powder charge, when a bullet is seated to the desired depth in the case neck. To do this file or cut a small groove lengthwise on the bearing surface of a bullet to be used, so that water will have a path to escape when the bullet is seated in a water-filled case. Take a resized case and make sure that a fired primer is seated in the case or something so that the water will not run or drip out. Weigh the case and grooved bullet together on your powder scale. Now, fill the case to the mouth with water, seat the bullet carefully to the desired seating depth, and make sure you dry off the case and bullet, and weigh the water-filled cartridge. Take the difference between the empty and full weights of that cartridge, with that particular bullet, and that seating depth and this is the cartridge capacity in grains of water.

Alternate method:

Take a resized case and make sure that a fired primer is seated in the case or something so that the water will not run or drip out. Weigh this case empty, and then fill the case carefully to the top, making sure there are no air bubbles still in side, without either a concave or convex meniscus in the water surface. Subtract the difference between the empty and full weights of that cartridge; record this as the full case capacity. You can drain the water out now. Carefully measure and record the case length. Next, measure the length of the bullet and establish the desired overall length of the loaded cartridge.

Seating depth can now be found by the formula: $S = C + B - L$ {where S is the seating depth (in inches), C is the case length (in inches), B is the bullet length (in inches), L is the overall length of the loaded cartridge (in inches)}.

Water displaced by seating a flat-base bullet. The amount can now be found by using S, don't worry we'll get to your boattails shortly, by the formula: $P = 198 * S * D^2$ {where P is the water displaced by the bullet (in grains), S is the seating depth (in inches), D is the bullet diameter, (in inches)}.

Water capacity of the powder chamber, of a flat-base bullet seated, can now be found by use the formula: $W = F - P$, where W is the water capacity of the cartridge (in grains), F is the full water capacity of the case (in grains), P is the water displaced by seating the bullet (in grains).

Let's work with these formulas for the time being. Let's say we have a .22-250 case and we're going to use the 55-grain SX bullet from Hornady. Empty, the case and bullet weighs 215.0 grains. The case full of water up to the case mouth and bullet weighs 259.6 grains. The difference in weight is 44.6 grains of water (F), case length is 1.911 inches (C), bullet length is .708 inches (B), and the desired overall length of the loaded cartridge is 2.415 inches (L).

To find the seating depth we use:

$$S = C + B - L$$

$$S = 1.911 + .708 - 2.415$$

$$S = 2.619 - 2.415$$

$$S = 0.204 \text{ seating depth in inches.}$$

For the water displaced by seating a flat-base bullet is:

$$P = 198 * S * D^2$$

$$P = 198 * 0.204 * 0.224^2$$

$$P = 198 * 0.204 * 0.050176$$

$$P = 198 * 0.010235904$$

$$P = 2.026708992 \text{ Water displaced by seating flat-base bullet in grains.}$$

We can now find the water capacity of the powder chamber of a flat-base seated cartridge by using:

$$W = F - P$$

$$W = 44.6 - 2.026708992$$

$$W = 42.573291008 \text{ in grains of water.}$$

Now were going to work with a boattail bullet for our .22-250. The bullet is a Sierra 55-grain BTHP. Empty, the case and bullet weighs 215.0 grains. The case full of water up to the case mouth and bullet weighs 259.6 grains. The difference in weight is 44.6 grains of water (F), case length is 1.911 inches (C), but now the bullet length is .720 inches (B), and the desired overall length of the loaded cartridge is 2.415 inches (L).

To find the seating depth we use:

$$S = C + B - L$$

$$S = 1.911 + 0.720 - 2.415$$

$$S = 2.631 - 2.415$$

$$S = 0.216 \text{ seating depth in inches.}$$

For the water displaced by seating a flat-base bullet is:

$$P = 198 * S * D^2$$

$$P = 198 * 0.216 * 0.224^2$$

$$P = 198 * 0.216 * 0.050176$$

$$P = 198 * .010838016$$

$P = 2.145927168$ in grain of water. Because were using a boattail bullet we are going to use the boattail correction formula.

Boattail bullet correction can be made quite easily, although this correction is usually less than one grain and hardly necessary. But, we crave math formulas so we're going to use it anyway.

The formula is: $K = 66 * H * (2 * D^2 - D * J - J^2)$ {where K is the correction for the boattail (in grains), H is the height (axial length) of the boattail (in inches), D is the bullet diameter (in inches), J is the tail diameter (small end) of the boattail (in inches). Add the correction (K) to the water capacity of the cartridge (W), which then becomes Water displaced by seating a boattail bullet: $W = F - P + K$ {where W, F, P, and K are found as described above}. From:

$$K = 66 * H * ((2 * D^2) - (D * J) - J^2) \text{ we have:}$$

$$K = 66 * 0.081 * ((2 * .224^2) - (0.224 * 0.202) - 0.202^2)$$

$$K = 66 * 0.081 * ((2 * 0.050176) - (0.224 * .202) - .040804)$$

$$K = 66 * 0.081 * (0.100352 - 0.045248 - 0.040804)$$

$$K = 66 * 0.081 * 0.0143$$

$$K = 66 * 0.0011583$$

$$K = 0.0764478 \text{ less grains of water compared to a flat-base bullet.}$$

For boattail bullets the formula for water capacity of cartridge powder chamber in grains now becomes:

$$W = F - P + K$$

$$W = 44.6 - 2.145927168 + 0.0764478$$

$$W = 42.454072832 + 0.0764478$$

$W = 42.530520632$ is the water capacity in grains of water of this boattail seated in this cartridge.

Bullet travel:

Bullet travel is determined by the distance that the base of the bullet must travel from the seated position in the cartridge until it clears the muzzle. Adding the length of the barrel (from the bolt face to the muzzle) to the seating depth of the bullet and subtracting from this sum the case length does this. My own .22-250 has a barrel length of 24.094 inches (E). We'll use are boattail bullet with a seating depth of 0.216 inches (S) and the case length has not changed, it is still 1.911 inches (C).

The formula would look like this:

$T = E + S - C$ where T is the bullet travel (in inches), E is the barrel length (in inches), S is the seating depth (in inches), and C is the case length (in inches).

$$T = 24.094 + 0.216 - 1.911$$

$$T = 24.310 - 1.911$$

$$T = 22.399 \text{ inches of bullet travel.}$$

Expansion ratio:

Expansion ratio is the ratio of the volume of the bore plus the powder chamber to the volume of the powder chamber alone. The volume is commonly expressed in cubic inches. The volume of the powder chamber is found by dividing the water capacity (in grains) by 252.4, the density of water at 70° F.

This formula would look like this:

$U = W \div 252.4$ where U is the volume of powder chamber (in cubic inches), W is the water capacity of the powder chamber (in grains), of our boattail.

$$U = 42.377625032 \div 252.4$$

$$U = 0.167898672868463 \text{ cubic inches of powder chamber volume.}$$

Now we find the effective volume of the bore. This is somewhere between groove diameter and the land diameter. This depends on the actual height, width, and number of the lands. You can find this out by having a casting done of a small section of the barrel and taking the measurements from the casting and finding out the effective area and multiply that by the bullets travel.

Or this can be estimated using this formula:

$Q = 0.773 * T * D^2$ where Q is the effective bore volume (in cubic inches), T is the bullet travel (in inches), D is the bullet diameter, and .773 is a barrel groove diameter constant (in inches).

$$Q = 0.773 * 22.399 * 0.224^2$$

$$Q = 0.773 * 22.399 * 0.050176$$

$$Q = 0.773 * 1.123892224$$

$$Q = 0.868768689152 \text{ cubic inches of effective bore volume.}$$

We now have enough to calculate the expansion ratio from the formula: $R = (Q + U) \div U$ where R is the expansion ratio, Q is the effective bore volume, and U is the volume of powder chamber.

$$R = (0.868768689152 + 0.167898672868463) \div 0.167898672868463$$

$$R = 1.03666736202046 \div 0.167898672868463$$

$$R = 6.17436305069916 \text{ expansion ratio.}$$

Estimating the powder charge:

We now have everything we need to find a powder type and estimate the powder charge. There are two different limitations on the amount of powder that can be used in a cartridge. The first the amount of powder a case will hold without compressing it based on its cubic capacity. The second is the amount of any particular powder that can be burned without producing excessive chamber pressure. What we are looking for is a powder that will fill the powder space of the cartridge, with a charge weight that closely

approaches, but not exceeds, the safe working chamber pressure. In interior ballistics "load density" is the ratio of the weight of the powder charge to the weight of water required to fill the volume of the powder chamber.

For DuPont IMR powders the maximum charge weight that can be loaded and still not compress the powder, is about 80% to 90% of the weight of water required to fill the powder space. To estimate our charge weight (I) the following formulas are used:

For IMR-4227 and IMR-4198:

$$I = 0.80 * W$$

and for all other IMR Powders:

$I = 0.86 * W$; where I is the charge weight (in grains), W is the water capacity of the powder chamber (in grains).

For IMR of 4227 and 4198 is:

$$I = 0.80 * 42.377625032$$

$$I = 33.9021000256 \text{ or } 33.9 \text{ grains of powder.}$$

Now for all other IMR powders is:

$$I = 0.86 * 42.377625032$$

$$I = 36.44475752752 \text{ or } 36.4 \text{ grains of powder.}$$

Powder characteristics:

The relative quickness is an arbitrary scale of DuPont powders. IMR-4350 has a value of 100, faster powders have higher numbers and slower powders have lower numbers.

Powder Type	Relative Quickness
IMR-4227	180
IMR-4198	160
IMR-3031	135
IMR-4064	120
IMR-4895	115
IMR-4320	110
IMR-4350	100
IMR-4831	95

This data is based on laboratory test instead of gun firings. The degree of relative quickness to meet our conditions depends mostly on three factors: 1). The ratio of powder-charge weight to bullet weight, sometimes called the mass ratio; 2). The sectional density of the bullet; and 3). The maximum working chamber pressure that we are prepared to accept. For our purpose we will establish a maximum working pressure of

45,000 c.u.p. to 50,000 c.u.p. Select the IMR powder that will be suitable and use one of the two formulas above.

Let say we want to use a powder in the neighborhood of IMR-4064 then we would use $I = 0.86 * 42.377625032 = 36.44475752752$ or 36.4 grains of powder for our .22-250 with a Sierra 55-grain BTHP. But this is not set in stone yet. Having found the estimated charge weight lets continue on.

Powder-selection Index:

This next step is to find the ratio of charge weight to bullet weight, which is also called the "mass ratio" for the load.

This is:

$A = I \div G$; where A is the mass ratio, I is charge weight (in grains), G is bullet weight (in grains).

$$A = 36.4 \div 55$$

$$A = 0.6618181818182 \text{ or } 0.66182.$$

We can find the sectional density of the bullet. Most reloading manuals give this to you but it is easily found by:

$Z = G \div (7000 * D^2)$; where Z is sectional density (in lb./in²), G is bullet weight (in grains), D is bullet diameter (in inches).

$$Z = 55 \div (7000 * 0.224^2)$$

$$Z = 55 \div (7000 * 0.050176)$$

$$Z = 55 \div 351.232$$

$$Z = 0.15659165451895 \text{ or } 0.157.$$

The next step is to find a number X that we will call the Powder-selection Index. The formula for finding this Powder-selection Index is:

$X = 20 + 12 \div (Z * \text{SQUARE ROOT of } A)$; where X is the Powder-selection Index, Z is sectional density (in lb./in²), A is mass ratio (charge weight $\div$ bullet weight).

$$X = 20 + 12 \div (0.15659165451895 * \text{SQUARE ROOT of } 0.6618181818182)$$

$$X = 20 + 12 \div 0.127390769185024$$

$$X = 20 + 94.1983479397246$$

$X = 114.1983479397246$ now round to the nearest whole number, 114 this is X. Compare this number to the Powder-selection Index Table below.

NOTE: If you used 0.86 in the equation $I = 0.86 * W$ then X needs to be between 145 to 81. If not then you must recalculate (I) using 0.80. But, our X was 114 indicating from the Powder-selection Index Table below that a powder is similar to IMR-4064, IMR-4895, IMR-4320 should be used. If X were say (Powder-selection Index below {X})

"Powder-Selection Index"

Powder Selection Index (X)	Powder indicated
	Powder indicated is much "slower" than IMR-4831.
Less than 81	There is no very suitable IMR canister powder available. Use only loads specifically pressure-tested, and do not experiment.
81 to 91	Powder indicated is "slower" than IMR-4831 and IMR-4350, and calculated charges must be reduced by about 5% to 10% if IMR-4831 or IMR-4350 is used.
91 to 110	Powder indicated is similar to IMR-4831 and IMR-4350.
110 to 125	Powder indicated is similar to IMR-4064, IMR-4895, and IMR-4320.
125 to 145	Powder indicated is similar to IMR-3031.
145 to 165	Powder indicated is similar to IMR-4198.
165 to 180	Powder indicated is similar to IMR-4227.
More than 180	Powder indicated is "faster" than IMR-4227. For lower pressures, IMR-4227 may be used, but velocities will be less than predicted.

Calculating the Velocity Performance:

Homer S. Powley's equation for predicting the performance of rifle loads using DuPont IMR powders. Here is the equation broken down into simpler sections:

$$M = (1 \div \text{the fourth root of } R)$$

$$N = (1 - M)$$

$$Y = (G + (I \div 3)).$$

This is:

$$M = 1 \div \text{the fourth root of } 6.17436305069916$$

$$M = 1 \div 1.57633326536771$$

$$M = 0.634383617963383, \text{ and}$$

$$N = (1 - 0.634383617963383)$$

$$N = 0.365616382036617, \text{ and}$$

$$Y = 55 + (36.44475752752 \div 3)$$

$$Y = 55 + 12.1482525091733$$

$$Y = 67.1482525091733.$$

We now put these variables in Powley's equation for muzzle velocity (V) and we have: (Oh, by the way, SQR means "the square root of".)

$$V = 8000 * \text{SQR} (I * N \div Y)$$

$$V = 8000 * \text{SQR} (36.44475752752 * 0.365616382036617 \div 67.1482525091733)$$

$$V = 8000 * \text{SQR} (36.44475752752 * 0.365616382036617)$$

$$V = 8000 * \text{SQR} 0.198438528085199$$

$$V = 8000 * 0.445464395979296$$

$$V = 3563.71516783437 \text{ or } 3564 \text{ fps.}$$

This is a very close estimate, my Hornady third edition reloading manual gives a 55-grain bullet with 35.4 grains of IMR-4064 lists at 3600fps, which is always faster than I get out of the manuals. So, I would be happy with this.

You can easily find the estimated changes in barrel length, cartridge capacity, or heavier or lighter bullet weight, ... etc. by changing one or more of the parameters and recalculating.

Load Density:

This is the ratio of charge weight to the capacity of the powder chamber (in grains of water). Varying either one changes the load density. The higher load density will produce higher chamber pressure and vice versa. This change in chamber pressure is approximately proportional to the square of the load density.

Let's say we have some commercial .30-06 cases to load. The cases hold 65.3 grains of water (W) and the charge weight is 57.5 grains of IMR-4350 (I).

This is:

$$LD = I \div W$$

$$LD = 57.5 \div 65.3$$

LD = 0.880551301684533 or 0.881. This charge is listed as about 50,000 c.u.p.

Now, Suppose we acquired some military brass and they held only 62.0 grains of water. What would happen if we used the same loads?

$$LD = I \div W$$

$$LD = 57.5 \div 62.0$$

LD = 0.927419354838710 or 0.927 an increase of 0.046868053154177 or 0.047 load density. So what you say? Well were not finished yet.

Pressure:

$$P = 50,000 * 0.927419354838710^2 \div 0.880551301684533^2$$

$$P = 50,000 * 0.860106659729448 \div 0.775370594898325$$

$$P = 50,000 * 1.10928459937565$$

$$P = 55,464.2299687825 \text{ or } 55,464 \text{ c.u.p.}$$

To make this max charge safe we'll need to reduce the load. But by how much is the question? We have the max pressure we want is 50,000 c.u.p. and we have the load density with the commercial cases is 0.881, now we just plug in the numbers and switch it around a bit.

0.880551301684533 = @ ÷ 62.0; so now we have:

$$@ = 0.880551301684533 * 62$$

@ = 54.5941807044410 or 54.6 grains of IMR-4350 to equal the same pressure as the commercial cases.

Calculating chamber pressure:

Chamber pressure like the entire section is for DuPont IMR POWDERS ONLY. Can chamber pressure be calculated based on cartridge characteristics and measurements that are easily measurable? The answer is a qualified yes, provided the limitations of the method are fully understood. From Powley's equation for calculating chamber pressure and we will present these equations in smaller sections so that they can be easier to handle. Find F2 from the table below and the muzzle velocity of the load must be known. The best way is to use an accurate chronograph, rather than by previous calculation.

We can define the three new variables. Let's call them K1, K2, and K3.

$$K1 = 0.0142 * I * F2 * V^2,$$

$$K2 = (0.53 * (G \div I)) + 0.26, \text{ and}$$

$$K3 = W * (R - 1.0).$$

Powley's equation for chamber pressure (P), in psi as measured by the crusher-type gage, is:

$$P = K1 * K2 \div K3.$$

F2 TABLE

Expansion Ratio	Mass Ratio (Charge Weight/Bullet Weight) (A)								
(R)	.20	.30	.40	.50	.60	.70	.80	.90	1.00
5.0	1.27	1.26	1.25	1.24	1.22	1.21	1.19	1.18	1.17
5.2	1.31	1.30	1.29	1.28	1.26	1.25	1.23	1.22	1.20
5.4	1.35	1.34	1.33	1.32	1.30	1.28	1.26	1.25	1.24
5.6	1.39	1.38	1.36	1.35	1.33	1.31	1.30	1.28	1.27
5.8	1.43	1.42	1.40	1.39	1.37	1.35	1.34	1.32	1.31
6.0	1.47	1.46	1.44	1.43	1.41	1.39	1.38	1.36	1.35
6.2	1.50	1.49	1.48	1.46	1.45	1.43	1.41	1.39	1.38
6.4	1.54	1.53	1.52	1.50	1.48	1.46	1.45	1.43	1.41
6.6	1.58	1.57	1.55	1.53	1.52	1.50	1.48	1.46	1.45
6.8	1.62	1.60	1.59	1.57	1.56	1.54	1.52	1.50	1.48
7.0	1.66	1.64	1.63	1.61	1.59	1.57	1.56	1.54	1.52
7.2	1.70	1.68	1.66	1.64	1.63	1.61	1.59	1.57	1.55
7.4	1.73	1.72	1.70	1.69	1.67	1.65	1.62	1.60	1.59
7.6	1.77	1.76	1.74	1.72	1.70	1.68	1.66	1.64	1.62
7.8	1.81	1.79	1.77	1.76	1.74	1.72	1.70	1.67	1.65
8.0	1.85	1.83	1.81	1.79	1.77	1.75	1.73	1.71	1.69
8.2	1.88	1.86	1.85	1.83	1.81	1.79	1.76	1.74	1.72
8.4	1.92	1.90	1.88	1.86	1.84	1.82	1.80	1.78	1.75
8.6	1.96	1.94	1.92	1.90	1.88	1.86	1.83	1.81	1.79
8.8	1.99	1.97	1.95	1.93	1.91	1.89	1.87	1.84	1.82
9.0	2.03	2.01	1.99	1.97	1.94	1.92	1.90	1.87	1.85
9.2	2.07	2.05	2.02	2.00	1.98	1.96	1.94	1.91	1.88
9.4	2.10	2.08	2.06	2.04	2.02	2.00	1.97	1.94	1.92
9.6	2.14	2.12	2.09	2.07	2.05	2.03	2.00	1.97	1.95
9.8	2.17	2.15	2.13	2.11	2.08	2.06	2.04	2.01	1.98
10.0	2.21	2.18	2.16	2.14	2.12	2.10	2.07	2.04	2.02
10.2	2.25	2.22	2.20	2.17	2.15	2.13	2.10	2.07	2.05
10.4	2.28	2.25	2.23	2.21	2.18	2.16	2.14	2.11	2.08
10.6	2.32	2.29	2.27	2.25	2.22	2.19	2.17	2.14	2.11

10.8	2.35	2.33	2.30	2.27	2.25	2.23	2.20	2.17	2.14
11.0	2.39	2.36	2.33	2.30	2.28	2.25	2.23	2.20	2.17
11.5	2.47	2.44	2.42	2.39	2.36	2.33	2.31	2.28	2.25
12.0	2.56	2.53	2.50	2.47	2.44	2.41	2.39	2.36	2.33
13.0	2.73	2.70	2.66	2.63	2.60	2.57	2.54	2.52	2.48

Let us do an example chamber-pressure. We have a 22-250 using a 55-grain bullet and 35.4 grains of IMR-4064 and our muzzle velocity is 3670 fps out of a 24-inch barrel. $I = 36.44475752752$ grains, $W = 42.377625032$ in grains of water, $G = 55$ grains, $R = 6.17436305069916$, $A = 0.661818181818182$, and $V = 3670$ -fps. We now find $F2$ to be 1.432 (after interpolation of (A) as being close to .66 and (R) as being close to 6.17).

$$K1 = 0.0142 * I * F2 * V^2$$

$$K1 = 0.0142 * 36.44475752752 * 1.432 * 3670^2$$

$$K1 = 0.0142 * 36.44475752752 * 1.432 * 13468900$$

$$K1 = 0.0142 * 36.44475752752 * 19287464.8$$

$$K1 = 0.0142 * 702926977.956577$$

$$K1 = 9981563.08698339$$

$$K2 = 0.53 * (G \div I) + 0.26$$

$$K2 = 0.53 * (55 \div 36.44475752752) + 0.26$$

$$K2 = 0.53 * 1.50913337696015 + 0.26$$

$$K2 = 0.799840689788878 + 0.26$$

$$K2 = 1.05984068978888$$

$$K3 = W * (R - 1.0)$$

$$K3 = 42.377625032 * (6.17436305069916 - 1.0)$$

$$K3 = 42.377625032 * 5.17436305069916$$

$$K3 = 219.277217141965$$

$$P = K1 * K2 \div K3$$

$$P = 9981563.08698339 * 1.05984068978888 \div 219.277217141965$$

$$P = 9981563.08698339 * 0.004833337013315$$

$$P = 48,244.2583190515\text{-psi (copper crusher)}.$$

A comparison was made between pressures calculated by this method, and pressures measured for the same loads as recorded in the DuPont handloader's guide for Smokeless Powders (1975-1976). The results indicated that the calculated pressure for cartridges from about .25-caliber to .338-caliber were usually within about 10% of the measured pressures, about equally divided above and below. For the smaller calibers, the calculated pressures were typically somewhat below the measured pressures, the average being about 4% lower for the 6mm/.243 calibers, and about 8% for the .22 calibers. For calibers larger than the .338, the calculated pressures were typically somewhat higher than the measured pressures, the average being about 7% higher for the .35-caliber

cartridges, and about 15% higher for the .458 Winchester Magnum. The .375 H&H Magnum, is the exception to the trend, gave calculated pressures in very good agreement with the measured pressures.